

A uniqueness theorem for the container derivative

∂ is the only nontrivial representable tangent structure
on finite-support polynomial functors

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over $\mathbb{N}$, per a referee report)

Abstract

A companion note showed that the Abbott–Altenkirch–Ghani–McBride one-hole derivative ∂ is the differential combinator of a Cartesian differential category of polynomial functors, realised by the dual-number substitution $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$; that settles *existence*. This note settles *uniqueness*. The dual-number substitution is *base change by the Weil algebra* $W = \mathbb{N}[\varepsilon]/\varepsilon^2$; more generally a representable first-order tangent functor is base change $T = (-) \otimes_{\mathbb{N}} D$ by an augmented square-zero $\mathbb{N}$ -algebra D (a finite Weil algebra over $\mathbb{N}$, with W the first instance), acting on finite-rank $\mathbb{N}$ -modules — equivalently, on the finite free $\mathbb{N}$ -modules that sit in **Poly** as the finite-support *linear* containers $S \cdot y$. We prove that among representable first-order tangent structures of this form there are *exactly two*, the trivial structure $T = \text{id}$ and the container derivative ∂ . The proof has two halves. The first is a rank rigidity: an infinitesimal object $D = y \oplus M$ has a *solid* infinitesimal part when the multiplication $\mu: M \otimes M \rightarrow M$ is invertible, and over the rig $\mathbb{N}$ the finite free modules are automatically free of a well-defined rank, so solidity reads $|S|^2 = |S|$, whose only finite solutions are 0 and 1. The second is the converse: every such tangent structure has a solid infinitesimal part. We prove this rig-internally from the universality of the vertical lift, replacing Lanfranchi–Lemay’s ring-theoretic short-exact-sequence step — which needs subtraction — by the observation that in the extensive category of $\mathbb{N}$ -modules the vertical bundle is a *direct coproduct*, so ranks *add*. The port is thus not only faithful but cleaner: no principal-ideal hypothesis and no subtraction. Crucially, on the linear layer base change $\otimes D$ coincides with container substitution $\triangleleft$ (and with the Dirichlet tensor), so the classification is at once a statement about Weil base change and about the container derivative — but it is *not* a statement about $\triangleleft$ -substitution on all of **Poly**, where there is no $\varepsilon^2 = 0$ truncation ($y^2 \triangleleft 2y = 4y^2$) and only $T = \text{id}$ survives. The finiteness is sharp for the rigidity: $M = \mathbb{N} \cdot y$ is a solid part of rank $|\mathbb{N}|$, since $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$; whether it underlies a genuine third tangent structure we leave open. The uniqueness answers, and sharpens, a question of Cockett (2012).

1 Introduction

From existence to uniqueness. In a companion note [11] we showed that the category **SPoly** of finitary polynomial functors is a Cartesian differential category whose differential combinator is the Abbott–Altenkirch–Ghani–McBride one-hole-context derivative ∂ [1], and that the attached

*Kodamai. Source and Lean certification: github.com/scotmacbeth/work-in-progress (revised 2026-10-05; the version reviewed at commit 57bac18 mis-stated the ambient category, see §2).

tangent bundle is computed by the oldest trick in differential algebra: substitute a dual number and read off the ε -coefficient,

$$T(p) = p(y + \varepsilon v) \bmod \varepsilon^2 = p(y) + \varepsilon v \cdot \partial p.$$

That note answers an *existence* question — there is a tangent structure whose vertical part is ∂ . The natural next question is *uniqueness*. The dual-number structure is one tangent structure; the identity functor $T = \text{id}$ is another (the trivial one). Are there others? Could there be an exotic first-order tangent bundle that is neither the derivative nor trivial? This note proves there cannot.

First we must say precisely in which category the question is asked — and here the right setting is not the one the formula $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$ first suggests. The $\varepsilon^2 = 0$ truncation is not substitution: it is *base change by the Weil algebra* $W = \mathbb{N}[\varepsilon]/\varepsilon^2$. The honest tangent functor is base change $T = (-) \otimes_{\mathbb{N}} D$ by an augmented square-zero $\mathbb{N}$ -algebra D (the dual numbers W being the first instance), acting on the finite-rank $\mathbb{N}$ -modules / finite Weil objects over $\mathbb{N}$ — the module layer on which Lanfranchi–Lemay’s representable tangent structures live, now over the rig $\mathbb{N}$ rather than a ring. It is *tempting*, but wrong, to read T as right-substitution $(-) \triangleleft W$ in the full substitution-monoidal category $(\text{Poly}, \triangleleft, y)$: full $\triangleleft$ has no $\varepsilon^2 = 0$ truncation ($y^2 \triangleleft 2y = (2y)^2 = 4y^2$, not $3y^2$; in general $y^n \triangleleft W = W^n$ is multiplicative), so on all of Poly the functor $(-) \triangleleft W$ is *not* the dual-number substitution and the only survivor is the trivial structure. The derivative survives precisely because the truncation — the Weil algebra — is present. What rescues the container reading is a coincidence (Lemma 5): on the *linear* layer $S \cdot y$, base change $\otimes D$, substitution $\triangleleft$, and the Dirichlet tensor all agree, and it is on this layer that the entire classification takes place.

Theorem 1 (Uniqueness). *Among representable first-order tangent structures $T = (-) \otimes D$ — base change by an augmented square-zero $\mathbb{N}$ -algebra $D = y \oplus M$ — on finite-rank $\mathbb{N}$ -modules there are exactly two: the trivial structure ($D = y$, $M = 0$, $T = \text{id}$) and the AAGM dual-number structure ($D = W = \mathbb{N}[\varepsilon]/\varepsilon^2$, $M = \langle \varepsilon v \rangle$, $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$, combinator ∂). Equivalently, on the linear layer of SPoly — where $\otimes D$ and $\triangleleft$ coincide — the container derivative is the only nontrivial representable first-order tangent structure.*

The question is Cockett’s. That polynomial functors should carry a differential structure is not a new hope. In an unpublished talk at FMCS 2012, *Can you Differentiate a Polynomial?*, Cockett [6] conjectured exactly this — that polynomial functors form a Cartesian differential, hence a tangent, category — fourteen years before the companion note proved it. But the 2012 slides are candid about how much was conjecture: the differentiation there is defined only on “separable” polynomials, closure under composition is asserted but not proved, and the chain rule is flagged as the hard open point (“*of course, need to prove this is a differential!! (chain rule is already challenging...)*”; “*Is this all completely sorted out? Absolutely not!*”). No uniqueness statement appears, nor could it: without the structure in hand there is nothing to count. The companion note supplied the structure; the present note supplies the count, and so closes the arc Cockett opened. (We flag one divergence in §9: combinatorial species, the setting nearest Cockett’s, are most naturally a tangent *bicategory* [10], structurally different from the SPoly one-category we work in.)

The template, and why it does not simply transfer. For commutative algebra the uniqueness question has a sharp answer. Lanfranchi and Lemay [9] classify the *representable* tangent structures on the opposite of the category of modules (equivalently, affine schemes): a representable tangent structure is a tensoring $T = D \otimes (-)$ by an *infinitesimal object* $D = R \oplus M$, and when the

base ring R is a principal ideal domain there are *exactly two* — the trivial one ($M = 0$) and the Kähler/dual-number one ($M = R$). Their argument has two load-bearing uses of ring structure. The *rigidity* half (a solid infinitesimal part has rank n with $R^{n^2} \cong R^n$, forcing $n^2 = n$, hence $n \in \{0, 1\}$; their Theorem 4.17, Corollary 4.18) needs R to be a PID, so that the infinitesimal module is free with a well-defined rank. The *converse* half (a tangent structure’s infinitesimal part *is* solid; their Theorem 4.8, via the universality of the vertical lift) computes a vertical bundle by a short exact sequence — i.e. by *subtraction* of ranks. Neither transfers to polynomial functors verbatim: the base there is the rig $\mathbb{N}$, which is not a PID and has no subtraction.

What we prove, and the one new idea. We prove Theorem 1 by porting *both* halves of Lanfranchi–Lemay to the Weil base-change setting over $\mathbb{N}$ — the tangent functor $T = (-) \otimes W$ on finite-rank $\mathbb{N}$ -modules — and in both the port turns out *cleaner* than its template. Throughout, the objects that carry the classification are *linear* ($S \cdot y$), the layer on which base change $\otimes W$, substitution $\triangleleft$ and the Dirichlet tensor coincide (Lemma 5); this is what licenses the container-derivative reading of the result.

- *Rigidity* (§4). An infinitesimal part is a *linear* module $M = S \cdot y$, forced by the additive-bundle axiom. Over $\mathbb{N}$ a linear module simply *is* a free module — the linear polynomial functors are by definition the free $\mathbb{N}$ -modules — so freeness is *automatic*, with no PID hypothesis. Solidity then reads $|S|^2 = |S|$, whose only finite solutions are 0 and 1: the finite solid parts are exactly $\{0, y\}$.
- *Converse* (§6). Every such tangent structure has a solid infinitesimal part. Here is the one new idea. The place Lanfranchi–Lemay use subtraction is the computation of the vertical bundle fibre P' by exactness. In the *extensive* category of $\mathbb{N}$ -modules that fibre is instead a *direct coproduct* of the surviving summands,

$$P' = y \oplus \varepsilon_{\text{out}} \oplus (M \triangleleft M),$$

so its rank is a *sum*, $1 + n + n^2$, computed by addition and never as a difference. The universality of the vertical lift (Cockett–Cruttwell [7]) identifies the second tangent bundle T_2 with the vertical bundle V ; evaluating this functor isomorphism at y gives $D_2 \cong P'$, i.e. $1 + 2n = 1 + n + n^2$, and the cancellative rig $\mathbb{N}$ (no zero divisors, *no subtraction*) forces $n^2 = n$, hence $n \in \{0, 1\}$.

Assembling the two halves with the forward correspondence (both surviving parts are realised — id and ∂) and representability [11] gives Theorem 1.

The finiteness is sharp for the rigidity. The finite-rank hypothesis cannot be dropped from the rank rigidity, and we exhibit the obstruction explicitly: $M = \mathbb{N} \cdot y$ is a solid infinitesimal *part*, because $M \otimes M = (\mathbb{N} \times \mathbb{N}) \cdot y \cong \mathbb{N} \cdot y$ by the Cantor pairing (§5, Proposition 10); and the converse’s rank equation $1 + 2\kappa = 1 + \kappa + \kappa^2$ is satisfied by every infinite cardinal κ , so neither half of the rank argument excludes it. This is the precise analogue of Lanfranchi–Lemay’s extra solid parts when finite generation is dropped (their Examples 4.20–4.21), and it tells us the load-bearing hypothesis is a *finite-dimensional* infinitesimal part — finitely many tangent directions — here taken as plain finiteness of the shape set. We are careful not to overclaim: $\mathbb{N} \cdot y$ is a solid part, but we have *not* verified that it assembles into a genuine third tangent structure, so whether uniqueness fails outright at infinite rank is left open (§5).

Certification. The arithmetic of both halves is machine-checked in a Mathlib-free Lean 4 development with no classical axioms: the idempotent collapse $n^2 = n \iff n \in \{0, 1\}$, the finite enumeration of solid ranks, the three coproduct-rank identities underwriting the “ranks add, never subtract” claim, and the universality collapse $1 + 2n = 1 + n + n^2 \iff n \in \{0, 1\}$ by additive cancellation. The Lean certifies the *arithmetic*; the structural inputs (the universality axiom, that $\triangleleft$ preserves pullbacks, that M is linear) are cited framework facts, and we are explicit about the boundary (§4, §6).

Contribution. The contributions are: the uniqueness theorem itself (Theorem 1), proved over a rig with neither a PID hypothesis nor subtraction; the rigidity half (finite solid parts = $\{0, y\}$); the rig-internal converse, whose single new idea is that extensivity turns the vertical bundle into a coproduct so that ranks add (§6); the technical observation that on the linear layer the base-change tensor $\otimes W$, the substitution tensor $\triangleleft$ and the Dirichlet tensor all coincide (Lemma 5), which is simultaneously what makes solidity computable in the bundle’s own tensor and what connects the Weil base-change classification to the container derivative; the sharpness of the finiteness hypothesis (Proposition 10); and the Lean certification of the arithmetic. The result is a clean deliverable for the tangent-category and Poly communities: *on the linear layer of finite-support polynomial functors — the finite free $\mathbb{N}$ -modules — the container derivative is the only nontrivial representable first-order tangent structure.*

Outline. §2 recalls containers, the substitution monoidal structure, infinitesimal objects, the representable tangent bundle and the universality of the vertical lift. §3 proves the three-tensor coincidence on the linear layer. §4 proves the rank rigidity (finite solid parts = $\{0, y\}$) and records its Lean certification. §5 exhibits the infinite-rank solid part. §6 proves the solidity converse from vertical-lift universality — the main new content — and records its Lean certification. §7 assembles the uniqueness theorem. §8 compares with Lanfranchi–Lemay. §9 concludes with the precedent and the sequel.

2 Preliminaries

We recall only what the argument uses; for the Cartesian differential and tangent structure on SPoly we refer to the companion note [11].

Containers and the substitution monoidal category. A *container in n variables* is a set S of *shapes* together with, for each $s \in S$ and each $i \in \{1, \dots, n\}$, a position set P_s^i ; its extension is the polynomial functor $\llbracket S \triangleleft P \rrbracket(\vec{X}) = \sum_{s \in S} \prod_{i=1}^n X_i^{P_s^i}$. We work with finitary containers, so only the cardinalities $|P_s^i|$ matter up to isomorphism [1]. Composition of containers is substitution of extensions, written $G \triangleleft F$, with unit the identity container y ; this makes $(\text{Poly}, \triangleleft, y)$ a (strict, once we pass to isomorphism classes) monoidal category, and the comonoids in it are exactly the small categories [2, 8]. A container is *linear* if it is of the form $S \cdot y$ for a set S , i.e. its extension is $\vec{X} \mapsto S \times X$ in one variable; *finite-support* if its shape set is finite. We write SPoly for the finite-support polynomial functors.

Infinitesimal objects and representable tangent bundles. We use the representable presentation of tangent structure of [9], transported along the dictionary $(\text{Mod}_R, \otimes_R, R) \rightsquigarrow (\text{Mod}_{\mathbb{N}}, \otimes_{\mathbb{N}}, \mathbb{N})$: the *same* module/algebra layer as Lanfranchi–Lemay, but over the rig $\mathbb{N}$ in place of a commutative ring. The tangent functor is *base change by a Weil algebra*, not substitution; the link to Poly is

that the finite free $\mathbb{N}$ -modules are exactly the finite-support linear containers $S \cdot y$, on which base change coincides with $\triangleleft$ (Lemma 5).

Definition 2 (Infinitesimal object / Weil algebra). An *infinitesimal object* is an augmented square-zero $\mathbb{N}$ -algebra $D = y \oplus M$: a pointed object with unit/zero $z: y \rightarrow D$ and augmentation $p: D \rightarrow y$, $p \circ z = \text{id}_y$, *infinitesimal part* $M = \ker(p)$ a finite free $\mathbb{N}$ -module with $M \cdot M = 0$ inside D (the $\varepsilon^2 = 0$ condition). The associated tangent functor is *base change* $T := (-) \otimes_{\mathbb{N}} D$, with tangent projection $A \otimes p: A \otimes D \rightarrow A$ and zero section $A \otimes z: A \rightarrow A \otimes D$. The dual numbers $W = \mathbb{N}[\varepsilon]/\varepsilon^2$ are the first such object; on a polynomial p , base change by W is the dual-number substitution $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2 = p(y) + \varepsilon v \cdot \partial p$, the $\varepsilon^2 = 0$ truncation being carried by W , *not* by composition.

Definition 3 (Representable first-order tangent structure). A *representable first-order tangent structure* is a tangent structure (in the sense of Cockett–Cruttwell [7], first order) whose bundle functor is base change $T = (-) \otimes D$ by an infinitesimal object $D = y \oplus M$. Its infinitesimal part M is called *solid* if the multiplication $\mu: M \otimes M \rightarrow M$ is an isomorphism.

Remark 4 (Convention: $\otimes$ versus $\triangleleft$, and why it matters). The tangent functor is base change $(-) \otimes D$, never substitution $(-) \triangleleft D$ on all of Poly. The distinction is real: full $\triangleleft$ has no $\varepsilon^2 = 0$ truncation (for $D = W$, $y^2 \triangleleft W = y^2 \triangleleft 2y = (2y)^2 = 4y^2 \neq 3y^2$, and $y^n \triangleleft W = W^n$ is multiplicative), so $(-) \triangleleft W$ is *not* the dual-number substitution and, as one checks, forces the trivial structure alone. What saves the container reading is Lemma 5: on the *linear layer* $S \cdot y$ — the only layer the rank arithmetic of §4–§6 ever touches — base change $\otimes$, substitution $\triangleleft$ and the Dirichlet tensor all agree. Every infinitesimal object below ($D = y \oplus M$, the second-order object D_2 , the vertical object P') is linear, so Lemma 5 lets us compute it by whichever of the three is convenient; we use the quick substitution formula $\triangleleft$ for these *object-level* computations while the tangent *functor* itself remains base change $(-) \otimes D$.

The dual numbers $W = \mathbb{N}[\varepsilon]/\varepsilon^2 = y \oplus \langle \varepsilon v \rangle$ are such an object: the augmentation is $\varepsilon \mapsto 0$ ($p: W \rightarrow y$), the zero is $y \hookrightarrow W$ (z), and base change $(-) \otimes W$ is the dual-number substitution T of [11] — the identity $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$ being base change, not $\triangleleft$ -substitution (Remark 4). Its infinitesimal part is the linear module $M = \langle \varepsilon v \rangle = 1 \cdot y$, of rank 1; the trivial structure is $D = y$, $M = 0$, $T = \text{id}$. That $T = (-) \otimes W$ is a genuine tangent structure — not merely an endofunctor — is the content of [11]; we take representability of ∂ as established there and ask only whether other infinitesimal objects give rise to further structures.

The universality of the vertical lift. Among the tangent-structure axioms of [7] we use one by name. The *second tangent bundle* is $T_2 := T \times_{\text{id}} T$ (the pullback of $T \xrightarrow{p} \text{id}$ along itself), and the *vertical bundle* V is the pullback of the tangent map $Tp: T^2 \rightarrow T$ along the zero $0: \text{id} \rightarrow T$. The *vertical lift* is a canonical comparison $T_2 \rightarrow T^2$, and the axiom of *universality of the vertical lift* ([7], part of the definition of a tangent structure) asserts that the square

$$\begin{array}{ccc} T_2 & \longrightarrow & T^2 \\ \downarrow & & \downarrow Tp \\ \text{id} & \xrightarrow{0} & T \end{array}$$

is a pullback; equivalently, that the induced map $T_2 \rightarrow V$ is an isomorphism. This is the single structural axiom the converse of §6 invokes.

Why the infinitesimal part is linear. For a tangent structure the projection $p: T \Rightarrow \text{id}$ is an *additive bundle*: each fibre carries a commutative-monoid structure and the bundle addition is linear. For $T = (-) \otimes W$ this forces the infinitesimal part M to be a *linear* module $M = S \cdot y$. A non-linear M — for instance εy^2 , whose ε -coefficient is $y^2 p'$ rather than a directional $v \cdot p'$ — has no vector-space fibre and fails the additive-bundle axiom. So linearity of M is *forced*, not assumed, and it is this linearity (not any projective-module theory) that supplies freeness in §4. We note that both the augmentation splitting $W = y \oplus M$ and the linearity of M use that we are in the *pointed* $\mathbb{N}$ -algebra setting (zero object, $0 = 1$ augmentation), which is exactly what base change over $\mathbb{N}$ provides and what full $(\text{Poly}, \triangleleft, y)$ does not; this is a second reason the classification is stated there and not in Poly at large.

3 Three tensors coincide on the linear layer

This is the linchpin of the whole note, and it carries two burdens at once. First, it reconciles the two readings of the tangent functor: the *bundle* is base change $\otimes W$, whereas the container derivative is phrased with substitution $\triangleleft$, and these are genuinely different operations on Poly (Remark 4). Second, the *solid* condition $M \otimes M \cong M$ lives in the “module layer” where rank multiplies, which looks like yet a third product. The following observation collapses all three onto the linear layer.

Lemma 5 (Three tensors coincide on the linear layer). *For linear modules, base change by a linear object, substitution, and the Dirichlet product all agree:*

$$(S \cdot y) \otimes (T \cdot y) = (S \cdot y) \triangleleft (T \cdot y) = (S \times T) \cdot y = (S \cdot y) \otimes_{\text{Dir}} (T \cdot y).$$

Proof. The extension of $S \cdot y$ is $X \mapsto S \times X$. Composing, $(S \cdot y) \triangleleft (T \cdot y): X \mapsto S \times (T \times X) = (S \times T) \times X = (S \times T) \cdot y$. The Dirichlet product of $\sum_s y^1$ and $\sum_t y^1$ is $\sum_{(s,t)} y^{1 \times 1} = (S \times T) \cdot y$. Base change of the free module $S \cdot y$ by the free module $T \cdot y$ tensors the generating sets, $(S \times T) \cdot y$. All three equal $(S \times T) \cdot y$. $\square$

Remark 6. The identity is special to the linear layer; off it the three products diverge. Substitution and Dirichlet already part on two-term polynomials ($(y+1) \triangleleft (y+1) = y+2$ while $(y+1) \otimes_{\text{Dir}} (y+1) = y+3$), and base change parts from substitution precisely at the $\varepsilon^2 = 0$ truncation: $y^2 \triangleleft (2y) = 4y^2$ whereas $y^2 \otimes W$ reads off $y^2 + 2\varepsilon v y$. So Lemma 5 is genuinely a statement about the linear layer, not a global identity — which is exactly why the uniqueness theorem is a statement about that layer. None of the three is the “doubling” Dirichlet action $(2y) \otimes_{\text{Dir}} (-)$, which sends $p \mapsto 2p$ and has nothing to do with the derivative.

Consequence. On the linear layer the bundle’s own base-change tensor $\otimes$ already multiplies rank, and agrees with $\triangleleft$, so the solid condition $M \otimes M \cong M$ reads $|S|^2 = |S|$ in the very tensor the bundle uses — and the entire rank arithmetic of §4–§6, taking place on this layer, may be read interchangeably in any of the three. This single coincidence is what makes the port faithful and what licenses the container-derivative reading of a result about Weil base change.

4 Rank rigidity: the solid parts are $\{0, y\}$

We classify the finite solid infinitesimal parts. Two structural facts about linear containers, then the rank computation.

Lemma 7 (Freeness is automatic over $\mathbb{N}$). *Every linear container $S \cdot y$ is the free $\mathbb{N}$ -module on the set S ; two linear containers are isomorphic iff their shape sets are equinumerous. The cardinal $|S| =: \text{rank}(M)$ is a complete isomorphism invariant.*

Proof. Under coproduct of containers and the $\mathbb{N}$ -action $n \cdot (S \cdot y) = (n \times S) \cdot y$, the linear containers form the free- $\mathbb{N}$ -module functor applied to sets: $S \cdot y$ is the $|S|$ -fold copower of y . A container isomorphism $S \cdot y \cong T \cdot y$ is, since morphisms between linear containers are just functions on shapes (positions being singletons), a bijection $S \cong T$. Hence rank is a complete invariant. $\square$

This is the analogue of Lanfranchi–Lemay’s “projective $\Rightarrow$ free $\Rightarrow$ rank is a complete invariant” (their Lemma 4.16, via Rotman’s structure theorem, which *requires* the base to be a PID). Here it is *automatic*: the linear polynomial functors *are* the free $\mathbb{N}$ -modules by definition, so no PID hypothesis is needed and the rigidity below holds over the rig $\mathbb{N}$ — strictly outside the Lanfranchi–Lemay hypotheses.

Lemma 8 (Solid $\Rightarrow$ idempotent rank). *If $M = S \cdot y$ is solid, i.e. $\mu: M \otimes M \rightarrow M$ is an isomorphism, then $|S|^2 = |S|$.*

Proof. By Lemma 5, $M \otimes M = M \triangleleft M = (S \times S) \cdot y$, of rank $|S|^2$ by Lemma 7. An isomorphism $\mu: (S \times S) \cdot y \cong S \cdot y$ forces, by Lemma 7, a bijection $S \times S \cong S$, i.e. $|S|^2 = |S|$. $\square$

Theorem 9 (Finite rigidity). *A finite-support solid infinitesimal part has rank in $\{0, 1\}$. Hence there are exactly two: $M = 0$ (rank 0) and $M = y$ (rank 1). By §2 these are the trivial structure and the AAGM dual-number structure, with combinator ∂ .*

Proof. For a natural number $n = |S|$ we have $n^2 = n \iff n(n-1) = 0 \iff n \in \{0, 1\}$, since $\mathbb{N}$ has no zero divisors. By Lemma 8 a finite solid part has $|S|^2 = |S|$, so $|S| \in \{0, 1\}$; by Lemma 7 the rank determines M , giving $M \in \{0, y\}$. $\square$

This is the Lanfranchi–Lemay rank computation $R^{n^2} \cong R^n \Rightarrow n^2 = n \Rightarrow n \in \{0, 1\}$, ported verbatim with the cardinality rank, and — per Lemma 7 — without their PID hypothesis.

Machine certification of the arithmetic. The finite arithmetic of Theorem 9 is certified in a Mathlib-free Lean 4 development with no classical axioms. The oracle proves: $n^2 = n \iff n \in \{0, 1\}$ on $\mathbb{N}$ (in general, **omega-free**); that the solid ranks in $0 \leq n \leq 12$ are exactly $\{0, 1\}$ (by **decide**); that these two ranks map through the structure assignment to exactly $\{\text{id}, \partial\}$, with the dual-number object of arity 2; that every rank $r \geq 2$ (the Whitney sums of §6) is non-solid; and that the Cantor pairing is injective on a 12×12 window, witnessing $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$ for §5. **#print axioms** on the packaged oracle reports only **propext** (a structural, non-classical axiom); the finite **decide** claims are individually axiom-free, with no **sorryAx**, **Quot.sound** or **Classical.choice**.

5 The finiteness hypothesis is sharp for the rigidity

Theorem 9 is stated for finite rank, and the hypothesis cannot be dropped from the rigidity: at infinite rank there is a third solid *part*.

Proposition 10 (Infinite-rank solid part). *The countable linear module $M = \mathbb{N} \cdot y$ is solid: $M \otimes M = (\mathbb{N} \times \mathbb{N}) \cdot y \cong \mathbb{N} \cdot y = M$, because $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$. It is a solid infinitesimal part distinct from 0 and y , so the rank rigidity (Theorem 9) fails without finiteness.*

Proof. Lemma 5 gives $M \otimes M = (\mathbb{N} \times \mathbb{N}) \cdot y$, and the Cantor pairing supplies a bijection $\mathbb{N} \times \mathbb{N} \cong \mathbb{N}$ (injectivity checked on a 40×40 window; see the Lean oracle of §4 for a certified 12×12 window). By Lemma 7 this is an isomorphism $\mu: M \otimes M \cong M$, and $\text{rank } M = |\mathbb{N}| \notin \{0, 1\}$. $\square$

Remark 11 (A solid part is not yet a tangent structure). We are deliberately careful about what Proposition 10 does and does not say. It exhibits a third solid *infinitesimal part*; it does *not* exhibit a third tangent *structure*. Passing from a solid part to a genuine tangent structure requires the forward direction of the correspondence $(\star)$ (§6), which we have verified only for the two finite primitives $\{0, y\}$ by appeal to [11]; we have *not* checked that $\mathbb{N} \cdot y$ carries the full Cockett–Cruttwell axioms. So whether *uniqueness* (as opposed to the rank rigidity) actually fails over all of Poly is left open. What is certain is that the finite-rank hypothesis is load-bearing for the rigidity computation, and this is all the uniqueness theorem below needs.

More generally every infinite cardinal κ satisfies $\kappa^2 = \kappa$ (under choice), so each yields a solid infinitesimal *part*. This is the exact analogue of Lanfranchi–Lemay’s extra solid parts when the finite-generation hypothesis is relaxed — their non-PID example $R = S \times S$ (Example 4.21) and non-finitely-generated $\mathbb{Z} \times \mathbb{Q}$ (Example 4.20). The moral is that the honest hypothesis is a *finite-dimensional* infinitesimal part: finitely many tangent directions, matching Lanfranchi–Lemay’s finite-generation (Niefield coexponentiability) condition, here realised as plain finiteness of the shape set. We will see in §6 that the converse’s rank equation does not exclude $\mathbb{N} \cdot y$ either, so finiteness is load-bearing for *both* halves of the rank argument (the question of a genuine third *structure* remaining open, Remark 11).

6 The solidity converse: universality of the vertical lift

Theorem 9 classifies the solid infinitesimal *parts*. To pass from parts to tangent *structures* we need the converse direction of the correspondence

$$(\star) \quad \begin{array}{l} \text{representable first-order tangent structure } T = (-) \otimes D \\ \text{part } M = \ker(D \rightarrow y). \end{array} \iff \text{solid infinitesimal}$$

The forward direction is immediate once the parts are known (§7). The converse — that every representable first-order tangent structure has a *solid* infinitesimal part — is Lanfranchi–Lemay’s Theorem 4.8, proved over a commutative ring, and is the step that did not transfer verbatim: their derivation computes a vertical bundle by a short exact sequence, hence by subtraction, which the rig $\mathbb{N}$ lacks. We now give a rig-internal proof. The engine is the universality of the vertical lift (§2); the one new idea is that, in the extensive category of $\mathbb{N}$ -modules, the vertical bundle is a *direct coproduct*, so its rank is computed by addition and never by a difference.

The entire argument lives within the linear layer (the finite free $\mathbb{N}$ -modules $S \cdot y$), where by Lemma 5 the tangent functor $T = (-) \otimes D$ *coincides* with substitution $(-) \triangleleft D$ — base change of a free module by a free module is substitution. We therefore compute the objects below with the quick substitution formula $\triangleleft$, which preserves pullbacks pointwise (Lemma 12); this is legitimate precisely because every object in sight is linear (it is for non-linear arguments, which never occur here, that $\otimes D$ and $\triangleleft D$ part company). Here $D = y \oplus M$, $M = S \cdot y$ linear of finite rank $n = |S|$ (linearity forced, §2).

6.1 Substitution preserves pullbacks on both sides

Lemma 12 ($\triangleleft$ preserves pullbacks). *For every $D \in \text{Poly}$ the functor $(-) \triangleleft D$ preserves pullbacks; and for every polynomial A the functor $A \triangleleft (-)$ preserves pullbacks.*

Proof. Pullbacks in $\mathbf{Poly}$ are computed pointwise. For $(-) \triangleleft D$: if $F = G \times_H K$ pointwise then for all X , $(F \triangleleft D)(X) = F(D(X)) = G(D(X)) \times_{H(D(X))} K(D(X)) = (G \triangleleft D \times_{H \triangleleft D} K \triangleleft D)(X)$, so $F \triangleleft D = G \triangleleft D \times_{H \triangleleft D} K \triangleleft D$. For $A \triangleleft (-)$: a polynomial functor A preserves connected limits ([8], Prop. 1.9), and a pullback is a connected limit, so A preserves the pointwise pullback $E_1(X) \times_{E_0(X)} E_2(X)$; hence $A \triangleleft (-)$ preserves pullbacks. $\square$

Consequence: the tangent data are all right-tensorings. The projection, zero, tangent map and the two bundles of $T = (-) \otimes D$ are $(-) \otimes$ (structure map on D) — which on the linear layer is $(-) \triangleleft (\text{structure map})$: $p = (-) \triangleleft \hat{p}$ with $\hat{p}: D \rightarrow y$ the augmentation and $z = (-) \triangleleft \hat{z}$ with $\hat{z}: y \rightarrow D$ the zero; and, by Lemma 12 applied so that $A \triangleleft (-)$ carries the relevant pointwise pullback,

$$T_2 = (-) \otimes D_2, \quad D_2 := D \times_y D; \quad V = (-) \otimes P', \quad P' := \hat{g}^{-1}(\text{im } \hat{z}), \quad \hat{g} := \hat{p} \triangleleft D: D \triangleleft D \rightarrow D.$$

All of D , D_2 , $D \triangleleft D$, P' are linear objects, and y is one of them.

6.2 The three coproduct counts

We work in the based setting: ε -grading modulo ε^2 within each copy of D , with the mixed term surviving across copies — the Weil-algebra model $\mathbb{N}[\varepsilon_1, \dots]/(\text{within-copy squares})$ that the base-change presentation of T [11] delivers. This is the content of “ $T = (-) \otimes D$ is base change,” and it is what makes D_2 (fibre product, no mixed term) differ from the composite $D \triangleleft D$ (mixed term present): the $\varepsilon^2 = 0$ truncation lives in D , not in the composition. In the extensive category of $\mathbb{N}$ -modules each object below is a direct coproduct of summands, so its rank is the *sum* of the summand ranks.

Lemma 13 (Coproduct ranks). *With $n = \text{rank } M$ and $\text{rank}(M \triangleleft M) = n^2$ (Lemmas 5, 7):*

$$\begin{aligned} D \triangleleft D &= y \oplus M \oplus M \oplus (M \triangleleft M), & \text{rank} &= 1 + n + n + n^2 = (1 + n)^2; \\ D_2 &= D \times_y D = y \oplus M \oplus M, & \text{rank} &= 1 + 2n; \\ P' &= y \oplus \varepsilon_{\text{out}} \oplus (M \triangleleft M), & \text{rank} &= 1 + n + n^2. \end{aligned}$$

Proof. ($D \triangleleft D$.) Since M is linear, $M(X) = S \times X$, so $(D \triangleleft D)(X) = D(D(X)) = X \oplus M(X) \oplus M(X) \oplus M(M(X))$: the base y , the inner copy’s M (ε_{in}), the outer copy’s M (ε_{out}), and the mixed $M \triangleleft M = (S \times S) \cdot y$. Ranks $1, n, n, n^2$.

(D_2 .) A fibre product over the base shares y and adjoins the two infinitesimal directions but creates *no* product of the two infinitesimals: $D_2 = y \oplus M \oplus M$, rank $1 + 2n$.

(P' .) The map $\hat{g} = \hat{p} \triangleleft D$ applies the augmentation $\hat{p}$ ($\varepsilon \mapsto 0$) to the *outer* copy: it kills ε_{out} and the mixed term $M \triangleleft M$ (both carry an outer- M factor) and fixes y and ε_{in} . The zero $\hat{z}$ has image the y -summand. Hence $P' = \hat{g}^{-1}(\text{im } \hat{z})$ is the sub-coproduct on which the ε_{in} -component vanishes: $P' = y \oplus \varepsilon_{\text{out}} \oplus (M \triangleleft M)$, rank $1 + n + n^2$. In the extensive category of $\mathbb{N}$ -modules this pullback *is* the sub-coproduct of surviving summands, so its rank is the sum $1 + n + n^2$ — not $\text{rank}(D \triangleleft D) - \text{rank}(\text{im } \hat{z})$. This is the step where Lanfranchi–Lemay form a difference of ranks (a short exact sequence over a ring), and where we do not. $\square$

Both $D_2 = (1 + 2n) \cdot y$ and $P' = (1 + n + n^2) \cdot y$ are linear (a coproduct $y \oplus (\text{free})$ is again linear), so Lemma 7 applies to them.

6.3 Universality forces $n^2 = n$

Theorem 14 (Solidity converse). *Let $T = (-) \otimes D$ be a representable first-order tangent structure on finite-rank $\mathbb{N}$ -modules, with $D = y \oplus M$ and $n = \text{rank } M$ finite. Then $n^2 = n$, hence $n \in \{0, 1\}$; i.e. M is solid and $M \in \{0, y\}$.*

Proof. By the universality of the vertical lift (§2), the comparison $T_2 \rightarrow V$ is an isomorphism of endofunctors, naturally in the argument. By Lemma 12 and its consequence, $T_2 = (-) \otimes D_2$ and $V = (-) \otimes P'$. Evaluate the functor isomorphism at the unit object y and use $y \otimes E = E$:

$$D_2 \cong P'.$$

Both are linear (Lemma 13), so by Lemma 7 their shape-counts agree:

$$1 + 2n = 1 + n + n^2.$$

Now use only that $\mathbb{N}$ is a cancellative rig with no zero divisors — *no subtraction, no additive inverses*. Additive left-cancellation of the common summand $1 + n$ gives $n = n^2$. For $n \geq 1$, multiplicative cancellation of n in $n \cdot n = n \cdot 1$ gives $n = 1$; and $n = 0$ satisfies $0^2 = 0$. Hence $n \in \{0, 1\}$, so $M \in \{0, y\}$ by Lemma 7, and $\mu: M \otimes M \rightarrow M$ is an isomorphism in each case. $\square$

Remark 15 (Why this is rig-internal). The subtraction that blocked a literal port lived in computing the vertical-bundle fibre P' by exactness. In Poly that fibre is a direct coproduct of the surviving summands (Lemma 13), so its rank is a *sum*. Everything downstream is additive/multiplicative cancellation in $\mathbb{N}$. The derivation never forms a difference of ranks, never inverts an element, and never constructs the solid map μ explicitly — solidity is *output*, read off from $n \in \{0, 1\}$. Lanfranchi–Lemay’s Theorem 4.8 instead builds μ and argues over a ring; the content we have replaced is precisely their one subtraction.

Remark 16 (The non-solid finite candidates are auxiliary bundles). The finite candidate objects excluded by Theorem 14 are the higher square-zero objects $D_r = y + \varepsilon_1 v_1 + \cdots + \varepsilon_r v_r$ ($\varepsilon_i \varepsilon_j = 0$, $r \geq 2$), with $M = \mathbb{N}^r$, $M \triangleleft M = \mathbb{N}^{r^2}$, $\text{rank } r^2 \neq r$: not solid. These D_r are precisely the Whitney sums $T_r = T \times_X \cdots \times_X T$, which occur in the tangent axioms as *auxiliary* bundles, not as standalone tangent structures. So the objects the converse rejects are exactly the ones that are not primitive structures — solidity coincides with indecomposability, as it should. (The non-solidity of every $r \geq 2$ is part of the certified Lean oracle below.)

Remark 17 (Consistency with sharpness). The rank equation $1 + 2n = 1 + n + n^2$ is satisfied by every infinite cardinal κ ($1 + 2\kappa = \kappa = 1 + \kappa + \kappa^2$), so universality does *not* exclude $M = \mathbb{N} \cdot y$. The converse is therefore as sharp as the rigidity: both hold only under finiteness, consistent with Proposition 10.

Machine certification of the arithmetic. The rig-internal arithmetic of Theorem 14 is certified in the same Mathlib-free, axiom-light Lean 4 style. The oracle proves the three coproduct-rank identities of Lemma 13 — $\text{rank}(D \triangleleft D) = (1 + n)^2$, $\text{rank}(D \triangleleft D) = \text{rank } P' + n$, $\text{rank}(D \triangleleft D) = \text{rank } D_2 + n^2$ — which are the honesty certificate for “ranks add, never subtract”; the universality collapse $\text{rank } D_2 = \text{rank } P' \iff n \in \{0, 1\}$, proved by `Nat.add_left_cancel` (literal additive cancellation, *no omega*) reducing to $n^2 = n \iff n \in \{0, 1\}$; the finite enumeration (survivors in $0 \leq n \leq 12$ are exactly $\{0, 1\}$); and the two witnesses $n = 1$ (universality holds, $\text{rank } D_2 = \text{rank } P' = 3$, the Weil rank of $\mathbb{N}[\varepsilon_1, \varepsilon_2]/(\varepsilon_1^2, \varepsilon_2^2, \varepsilon_1 \varepsilon_2)$) and $n = 2$ (excluded, $5 \neq 7$). `#print axioms`

on the packaged oracle reports only **propext**. The Lean certifies the *arithmetic collapse* and the additivity of the counts; it does *not* formalise the structural inputs — the universality axiom ([7]), that $\triangleleft$ preserves pullbacks (Lemma 12, via [8]), or that M is linear (the additive-bundle axiom) — which are cited framework facts, and we are explicit that they are the mathematical, not the decidable, content.

7 The uniqueness theorem

With both halves in hand the correspondence $(\star)$ is an equivalence, and the uniqueness theorem assembles.

Forward: solid $\Rightarrow$ structure. A finite solid part yields a tangent structure in the two surviving cases. $M = 0$ gives $T = \text{id}$; $M = y$ gives the AAGM dual-number structure, which *is* a tangent structure — the full Cartesian differential / tangent verification is the content of [11]. Since Theorem 9 leaves no third finite solid part, the forward direction never has to manufacture a third structure.

Theorem 18 (Uniqueness). *Among representable first-order tangent structures $T = (-) \otimes D$ (base change by an augmented square-zero $\mathbb{N}$ -algebra $D = y \oplus M$) on finite-rank $\mathbb{N}$ -modules there are exactly two: the trivial structure ($D = y$, $T = \text{id}$) and the AAGM dual-number structure ($D = W = \mathbb{N}[\varepsilon]/\varepsilon^2$, $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$, combinator ∂). On the linear layer of **SPoly**, where $\otimes D$ and $\triangleleft D$ coincide (Lemma 5), this says the container derivative is the only nontrivial such structure.*

Proof. Let $T = (-) \otimes D$ be such a structure, $D = y \oplus M$. By Theorem 14 its infinitesimal part is solid with $\text{rank } M \in \{0, 1\}$; by Theorem 9 (equivalently Lemma 7) this forces $M \in \{0, y\}$, hence $D \in \{y, W\}$ up to isomorphism. By the forward direction both are realised, by id and ∂ respectively. Hence exactly two. $\square$

This is Theorem 1, now unconditional: the conditional hypothesis of the earlier draft — that every representable first-order tangent structure has a solid infinitesimal part — is discharged by Theorem 14.

8 Comparison with Lanfranchi–Lemay

The theorem is a clean deliverable for two audiences. For the tangent-category community [7, 9] it exhibits a rig-based model in which the representable-tangent rigidity *and* its vertical-lift converse hold for strictly weaker reasons than over a PID — no principal ideals, no subtraction — isolating exactly which uses of ring structure in [9] were essential (none, for finite-dimensional infinitesimal parts) and which were incidental (the PID and the short exact sequence, both replaced). For the **Poly** and containers community [1, 2] it upgrades “the container derivative is a tangent structure” [11] to “it is the *only* nontrivial one,” a uniqueness theorem for a construction that is otherwise defined by a formula.

9 Conclusion: precedent and sequel

We have proved that among the representable first-order tangent structures given by Weil base change $T = (-) \otimes D$ on finite-rank $\mathbb{N}$ -modules there are exactly two, the trivial one and the container

Lanfranchi–Lemay (affine schemes, $\mathbf{Mod}_R$)	This note (Weil base change over $\mathbb{N}$)
tangent functor $T = D \otimes_R (-)$, base change	tangent functor $T = (-) \otimes_{\mathbb{N}} D$, base change ($= \triangleleft$ on the linear layer, Lemma 5)
base a commutative ring R ; rigidity needs R a PID	base the rig $\mathbb{N}$ (no subtraction); rigidity needs no PID
projective $\Rightarrow$ free = Rotman structure theorem (uses PID)	linear module <i>is</i> free $\mathbb{N}$ -module (Lemma 7) — automatic
rank in R -modules, $R^{n^2} \cong R^n$	cardinality rank, $(S \times S) \cong S$ (Lemmas 5, 7)
rigidity: solid $M \otimes_R M \cong M \Rightarrow n^2 = n \Rightarrow$ $n \in \{0, 1\}$	rigidity: solid $M \otimes M \cong M \Rightarrow S ^2 = S \Rightarrow$ $ S \in \{0, 1\}$
converse: vertical bundle fibre via short exact sequence (rank by subtraction)	converse: fibre P' a direct coproduct $y \oplus \varepsilon_{\text{out}} \oplus (M \otimes M)$ (rank by addition ; Lemma 13)
two structures: trivial + Kähler/dual-number extra solid parts at non-PID / non-f.g. (Ex. 4.20–4.21)	two structures: $\text{id} + \partial$ (Theorem 18) extra solid <i>part</i> at infinite rank $\mathbb{N} \cdot y$ (§5; not shown to be a structure)

Table 1: The port is faithful (same base-change tangent functor, same rank-idempotent mechanism, same vertical-lift converse) and sharper on two axes: freeness is automatic over $\mathbb{N}$ (rigidity over a rig, not a PID), and the vertical bundle is a coproduct (converse by addition, not subtraction). The three-tensor coincidence (Lemma 5), which identifies base change with container substitution on the linear layer, and the extensivity of $\mathbb{N}$ -modules (Lemma 13) are the linchpins specific to this setting.

derivative ∂ (Theorem 18); and because base change coincides with container substitution on the linear layer (Lemma 5), this is at the same time the statement that ∂ is the unique nontrivial representable first-order tangent structure on that layer of finite-support polynomial functors. The proof ports Lanfranchi–Lemay’s affine-scheme classification in both halves, and in both the rig $\mathbb{N}$ makes the port *cleaner*: freeness is automatic, so the rigidity needs no PID (Lemma 7); and the vertical bundle is a direct coproduct, so the converse needs no subtraction (Lemma 13, Theorem 14). Two technical observations carry the port: on the linear layer the base-change tensor $\otimes D$, the substitution tensor $\triangleleft$ and the Dirichlet tensor all coincide (Lemma 5), and $\mathbb{N}$ -modules are extensive so that ranks add. The arithmetic of both halves is Lean-certified with no classical axioms. The finite-rank hypothesis is sharp for the rigidity: $\mathbb{N} \cdot y$ is a third solid *part*, and neither half of the rank argument excludes it, so the honest hypothesis is a finite-dimensional infinitesimal part. Whether $\mathbb{N} \cdot y$ underlies a genuine third tangent *structure* — and hence whether uniqueness itself (not just the rigidity) needs finiteness — we leave open (Remark 11).

Precedent. The hope that polynomial functors carry a differential structure is Cockett’s. His unpublished FMCS 2012 talk *Can you Differentiate a Polynomial?* [6] conjectured that they form a Cartesian differential, hence a tangent, category — the thesis the companion note [11] later proved. The 2012 slides, though, are explicit that this was conjecture: the differentiation there is defined only on “separable” polynomials, closure under composition is asserted rather than proved, and the chain rule is named as the hard open point. (The slides even record, as an aside, that this is “a differential category in which negation is unnatural” — the same absence of additive inverses that we have just turned to advantage, since it is exactly what makes the vertical-bundle coproduct,

rather than a short exact sequence, the right tool.) No uniqueness statement appears in 2012. The present note supplies it, and so answers Cockett’s question in its strongest form: not only *can* you differentiate a polynomial, but differentiation is the *only* nontrivial way to do so. We flag one caveat, due to Cruttwell via [10]: combinatorial species — the setting closest to Cockett’s — are most naturally a tangent *bicategory*, with species as 1-morphisms, which is structurally different from the SPoly one-category treated here; our uniqueness is a statement about the latter.

Sequel. A rigid tangent structure is the entry condition for the Cartan-calculus / Lie-algebroid program of Aintablian and Blohmann [3], which asks when a tangent category supports a scalar “ring object” R making each TX an R -module compatibly with the tangent axioms, and thereby upgrades a tangent structure to a Lie-algebroid structure with a full Cartan calculus. That SPoly now has a *unique* nontrivial first-order tangent structure makes the question well-posed: is there such an R for (SPoly, ∂) ? The negation remark above is a warning sign — a scalar ring object requires additive inverses, which $\mathbb{N}$ does not have — and determining exactly how much of the Cartan calculus survives over a rig is the natural next problem. We leave it, together with higher Weil algebras $\mathbb{N}[\varepsilon]/\varepsilon^k$ ($k \geq 3$) and iterated tangents (first-order out of scope here, as for [1, 9, 5]), to future work.

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